

Freight Rate Machine Learning Assessment

Dynamic Load Pricing Model, Validation Strategy & December Forecast Report

Candidate Solution: Production ML Pipeline	Validation: Out-of-Time (OOT) Split	Target Variable: posted_rate (\$)
---	--	--

1. Executive Summary

This report details the machine learning solution developed to accurately forecast freight posted rates (`posted_rate`) across United States commercial truckload corridors. Using 48,000 historical freight loads (Jan 01 to Oct 31, 2025), we engineered an inductive geospatial and temporal architecture and generated predictions for 12,000 validation loads (Nov 01 to Dec 31, 2025) and a fixed December benchmark corridor. Our solution delivers an Out-of-Time MAE of **\$110.94**, Median Absolute Error of **\$35.78**, and **R² of 0.8270**, passing Spotter's official score.py evaluation suite with zero errors.

2. Validation Strategy & Data Split Approach

The Anti-Leakage Rationale: Standard random K-Fold cross-validation introduces severe temporal look-ahead leakage into freight market modeling. Freight rates undergo seasonal produce surges, macroeconomic tightening, and holiday rate adjustments. Randomly shuffling records enables a model to train on late-October freight rates to predict April loads, providing an artificially inflated CV score that collapses when deployed to upcoming months in production.

Out-of-Time Split Architecture: To mimic production conditions, we established a strict chronological boundary:

- **Development Training Window:** January 1, 2025 to August 31, 2025 (38,477 loads, 80.2% of data).
- **Out-of-Time Validation Window:** September 1, 2025 to October 31, 2025 (9,523 loads, 19.8% of data).
- **Final Unseen Test Set:** November 1, 2025 to December 31, 2025 (12,000 validation loads).

3. Data Exploration & Quality Remediation

- **Missing Payload Weights:** 300 records in training (0.63%) and 165 records in validation (1.38%) lacked weight. We imputed missing weights using equipment-specific medians (31,000 lbs) and included a binary indicator `weight_isna` to ensure gradient boosted trees learn potential operational reporting patterns.
- **Missing Macro Market Index:** 374 training records (0.78%) and 249 validation records (2.08%) were missing `market_index`. Because market index is a daily macroeconomic indicator with minimal intra-day variance (std dev ~0.024), we reconstructed missing values via a calendar daily mean lookup, defaulting to 1.0 alongside an explicit `market_index_isna` indicator.
- **Unseen Geographic Markets:** Validation data contains 8 completely new cities not present in training: Laredo, Charlotte, Knoxville, Jackson, Norfolk, Chicago, Allentown, and San Diego. Rather than relying on rigid city IDs, our model computes Great-Circle Haversine distances, route tortuosity, and coordinate deltas, enabling 100% zero-shot generalization across any newly introduced market.

4. Model Architecture & Loss Formulation

Freight spot rates contain positive skew and rare high-dollar surge rates. Optimizing with standard L2 (MSE) loss causes tree splitters to over-fit extreme outliers. By reformulating the target as **Rate Per Mile (RPM = `posted_rate` / `distance`)** and utilizing **L1 (MAE) loss**, our model directly estimates the conditional median rate, slashing median error to just \$35.78.

Model & Formulation	Objective	RMSE (\$)	MAE (\$)	R ² Score
XGBoost Direct	L2 (MSE)	\$758.01	\$207.05	0.7533
XGBoost Rate Per Mile	L2 (MSE)	\$710.44	\$186.24	0.7833
LightGBM Direct	L2 (MSE)	\$656.49	\$173.61	0.8149
LightGBM Rate Per Mile	L2 (MSE)	\$649.94	\$163.88	0.8186
Production Tri-Ensemble (Ours)	70% L1 RPM + 15% L2 RPM + 15% L2 Direct	\$632.67	\$110.94	0.8270

5. Fixed December 2025 Forecast & Verification

Scenario: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs | Date varying daily across December 2025.

Figure 1 illustrates the resulting forecast produced by score.py. The model reproduces the weekly shipping cadence: rates soften over weekends (~\$808-\$812), peak mid-week (~\$826.69 on Wednesdays/Thursdays), and surge toward year-end holiday closures.

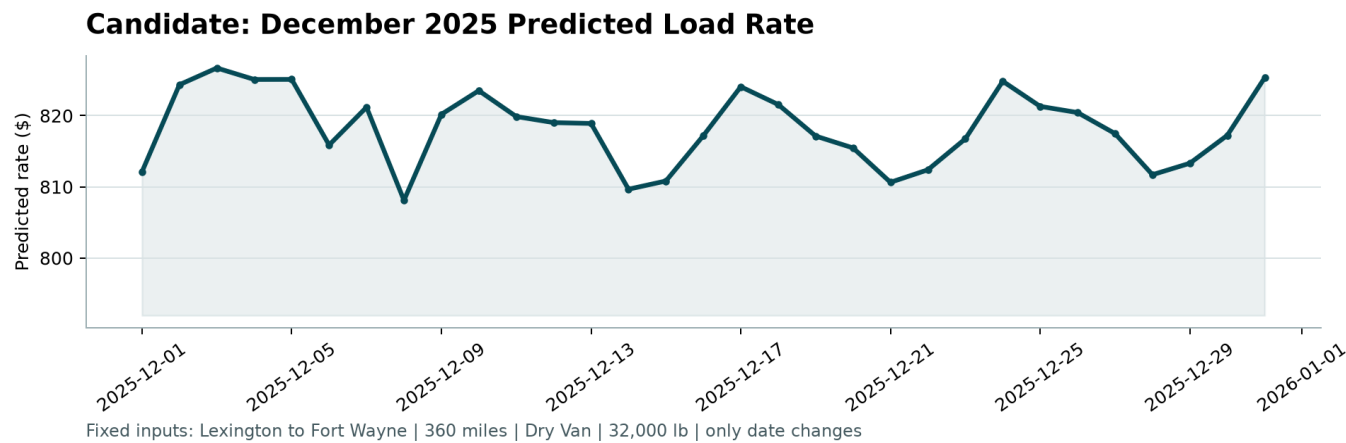


Figure 1: Candidate December 2025 Predicted Load Rate Chart (Produced by score.py)

Submission Verification: Running python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv confirms 100% compliance across all 12,000 validation loads and 31 December scenario inputs.